CONFAI 2026 · AI IN PRACTICE (EXTENDED TRACK) · POSTER EXHIBITION

PLAKSHA UNIVERSITY
ConfAI 2026 · AI in Practice

Literature Discovery with Multi-Signal Ranking, Citation-Grounded Hybrid RAG, and 2D Novelty Geometry

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GitHub code

Live demo

Keyword search returns unranked lists. General chat models answer without pointing to the passages that support them.

Dense MRR falls 41.9% from 10 to 5,663 papers. Hybrid RRF adds +0.105 MRR (95% CI 0.015 to 0.197).

1 Find papers
- arXiv catalog: 5,752 papers
- Citation graph: 101,716 nodes
- 4-signal ranker, 4 presets

20-60 papers

2 Ask questions
- User-gated PDFs: 553,516 chunks
- Dense (768d) + BM25 via RRF k=60
- Cited answers with [n] passages

5,707 scored

3 Judge novelty
- References: S2, OpenAlex, arXiv
- NNLS cone projection
- Outputs: m, Delta, Res

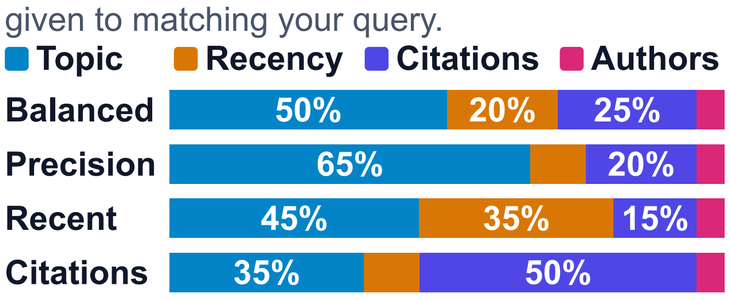
1 Find papers

Which papers should I read first?

Each paper gets a score from four signals: topic match, recency, citations, authors. You choose the mix.

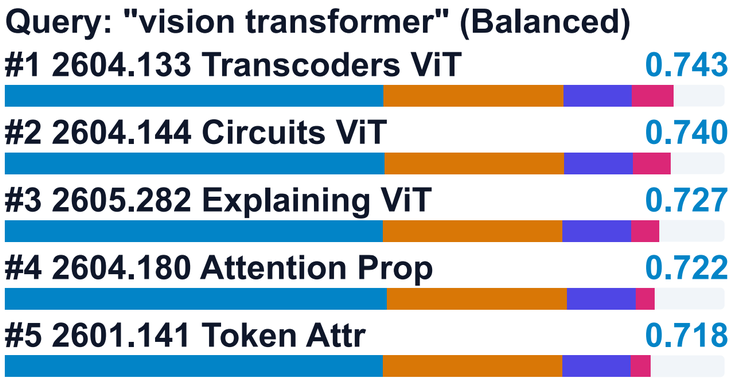
Presets trade topic match against recency

Each bar is the whole score. Blue is the share given to matching your query.



One query, five papers, and what drove each score

Coloured segments show how much each signal added to the score.



Changing the preset reorders the list

Number = rank, 1 is best. Read along a row to see one paper move.

Paper	Bal	Prec	Rec	Cit
Transcoders	1	1	1	4
Circuits	2	2	2	5
Explaining	3	3	3	6
Foundations	4	5	8	1
Survey	6	7	7	2

Citations preset lifts classic foundations (#1).

2 Ask questions

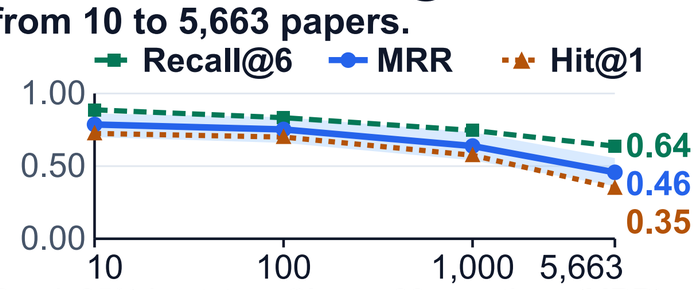
Can I trust answers from a big library?

You pick the papers. The system retrieves passages and answers with numbered citations. We tested how well.

Retrieval gets worse as the library grows

MRR: rank of right passage (1 = best). Hit@1: it is first. Recall@6: in top 6.

MRR falls 41.9% and Hit@1 51.7% from 10 to 5,663 papers.

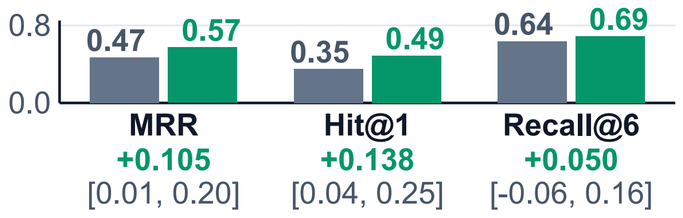


Band: 95% bootstrap CI over 80 questions (MRR).

Keyword matching ranks the right passage higher

Same 80 questions. Under each pair: gain from hybrid, with 95% interval.

Ranking gains hold, recall gain does not. Retrieval is 2.7x slower.



Citations are real, but the system over-refuses

Rows: question type. Columns: what it did. Green cells are correct.

19 of 80 answerable questions were wrongly refused.

Positive means should refuse	Refused	Answered
Out of scope	TP: 13	FN: 7
Answerable	FP: 19	TN: 61

Refusal recall 65.0% (CI 43 to 82)
Refusal precision 40.6% (CI 26 to 58)
Citations point to a retrieved passage 98.9%
Does not test if passage supports claim (89/90).

3 Judge novelty

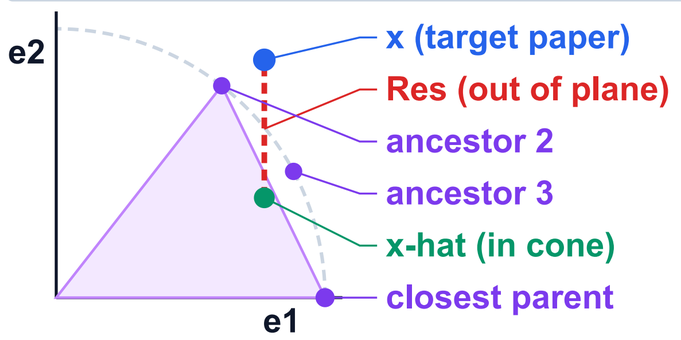
How original is a paper?

We rebuild a paper from the papers it cites. What cannot be rebuilt is the part that is new.

Novelty is measured against a paper's own references

Shaded cone: all blends of cited papers. x-hat: nearest blend to x. Res: the gap.

$x\text{-hat} = Y\lambda^* \ (\lambda \geq 0),\ \Delta = R^2_+ - m^2,\ \text{Res} = \sqrt{(1 - R^2_+)}$



5,176 of 5,707 papers (90.7%) land in High Synergy.

Two real papers: one close parent, one blend

m: similarity to closest parent. Delta: gain from blending. Res: new part.

BIG-bench used 80 of 706 references, so its Res is uncertain.

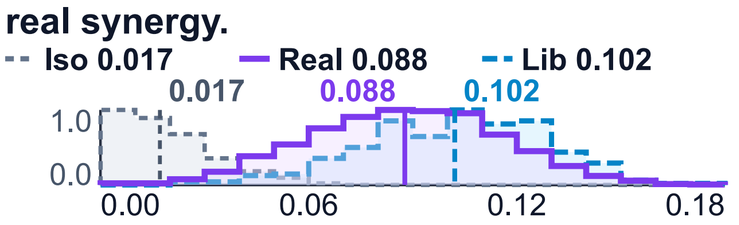
Paper	Refs	m	Δ	Res	Share
Grad-CAM++ [2]	18/18	0.94	0.05	0.27	49.7%
BIG-bench [3]	80/706	0.77	0.14	0.51	24.7%

Top parents: Grad-CAM (49.7%), BERT (24.7%).

Real citations score no higher than random ones

Curves: how often each value occurs. Random sets match each reference count.

So delta alone does not show real synergy.



Diff: -0.014 (95% CI -0.025 to -0.017).

Limitations

- Refusal precision is 40.6% (23.8% of answerable queries declined).
- Citation validity checks passage existence, not entailment.
- Gold set: 100 LLM questions on 10 papers, so added papers can score as misses.

Conclusions

- Dense MRR falls 41.9% and Hit@1 51.7% from 10 to 5,663 papers.
- Hybrid RRF adds +0.105 MRR over dense, CI 0.015 to 0.197.
- Cone novelty scores exceed an isotropic baseline (0.088 vs 0.017).

Future work

- Run hybrid retrieval across all sizes and test external novelty signals.
- In-person queries: gold generation, hybrid costs, reference coverage.
- Contact: Ajay Sharma, Dr. Subrat Panda (Plaksha)

References: [1] Cormack et al., Reciprocal Rank Fusion, SIGIR 2009. [2] Chattopadhay et al., Grad-CAM++, WACV 2018. [3] Srivastava et al., BIG-bench, arXiv 2206.04615, 2022.

Index sizes: 5,657 papers (hybrid and paired runs), 5,663 (scaling sweep). RRF k=60, alpha=0.5 and section priors are fixed defaults, not tuned on the gold set. CIs: bootstrap for retrieval, Wilson for proportions.